Example: Average # of defective per square yard of cloth is 1.2
And Poisson assumptions hold.

a) Find the probability of 1 defective in 1 square yard.

X = # of defective in 1 square yard.

$X \sim \text{Poisson}(\lambda = 1.2)$

$$P(X=1) = \frac{e^{-\lambda}\lambda^{x}}{x!} = \frac{e^{-1}\,1.2^{1}}{1!} = 0.3614.$$

b) Find the probability of 2 defective in 2 squared ~~yo~~ yards.

Y = # of defective in 2 square yards.

$Y \sim \text{Poisson}(\lambda = 1.2 \times 2 = 2.4)$

$$P(Y=2) = \frac{e^{-2.4}\,2.4^{2}}{2!} = 0.2613.$$

Continuous R.V

Recall: A continuous R.V can assume any value in an interval.

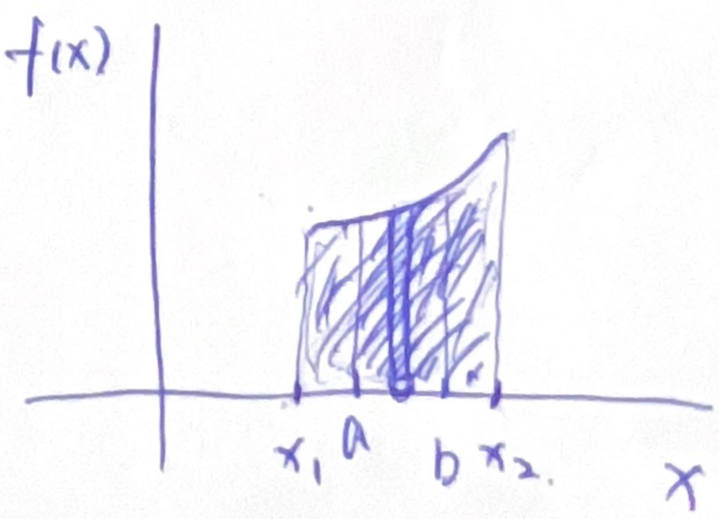
Def'n: The probability density function (pdf) describes the probabilistic behavior of a continuous R.V.

The pdf satisfies.

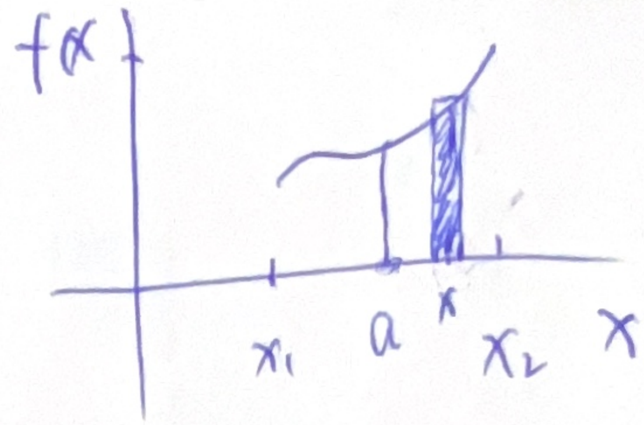
1) $f(x) \geq 0 \quad \forall x \in \mathbb{R}$.

2) If $a \leq b$, then $P(a \leq X \leq b) = \int_a^b f(x)\,dx$.

3) $\int_{-\infty}^{+\infty} f(x)\,dx = 1$.



Note: 1) $P(X=a) = 0$



2) $P(a < X < b)$

$$= P(a \le X \le b) - \underbrace{P(X=a)}_{=0} - \underbrace{P(X=b)}_{=0}$$

$$= P(a \le X < b)$$

$$= P(a < X \le b).$$

3) The pdf is proportional to the instantaneous probability.

i.e., for <u>small ε</u>. $P(X - \frac{\varepsilon}{2} < X < X + \frac{\varepsilon}{2}) \approx \varepsilon f(X)$.